MTH314

Timed Online Assignment – January Semester 2022

Real Analysis II

Tuesday, 24 May 2022 **1:00 pm – 3:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Define a function

$$f(x) = \sqrt{x + \frac{1}{x}}.$$

Find the value of $f'(2)$ from first principles, that is, by using the limit definition.

(6 marks)

(b) Use a non-calculator method to prove that the function

$$g(x) = 3x^5 - 10x^3 + 75x - 68$$

is strictly increasing for all $x \in \mathbb{R}$, and find the value of $(g^{-1})'(0)$.

(7 marks)

(c) Define the function

$$h(x) = |\sin(\pi x)|\,.$$

Show that $h(x)$ is not differentiable at the points where x is an integer by computing the left and right derivatives of $h(x)$ at these points.

(5 marks)

Question 2

Let r be a positive real number. A function $f(x) : [0.5, 3.5] \longrightarrow \mathbb{R}$ is differentiable four times on the closed interval $[0.5, 3.5]$, and satisfies $f(2) = 1$, $f'(2) = 1/3$, $f''(2) = 5/18$, $f'''(2) = 10/27$, and $\left|f^{(4)}(x)\right| \leq r$ for all $0.5 \leq x \leq 3.5$.

(a) Find the Taylor polynomial $P_3(x)$ of $f(x)$ about the point $x = 2$, that is, the Taylor representation about $x = 2$ up to and including the term involving $(x-2)^3$.

(4 marks)

(b) Find the maximum error in estimating $f(x)$ in the interval $0.5 \leq x \leq 3.5$ using the Taylor polynomial $P_3(x)$ about $x = 2$. Express your answer in terms of r.

(3 marks)

(c) The function

$$g(x) = \frac{2^{2/3}}{\sqrt[3]{(4-x)^2}}$$

is infinitely differentiable on the closed interval $[0.5, 3.5]$, and satisfies $g(2) = 1$, $g'(2) = 1/3$, $g''(2) = 5/18$, $g'''(2) = 10/27$.

(i) For the case where $x_0 = 2.5$, calculate the error in estimating $g(x_0)$ using the Taylor polynomial $P_3(x)$ about $x = 2$, expressing your answer in the form

$$\frac{g^{(4)}(c)}{4!}(x_0 - 2)^4 ,$$

where c is a real constant to be determined to an accuracy of 2 decimal places.

(7 marks)

(ii) Use the Taylor polynomial $P_3(x)$ of $g(x)$ about the point $x = 2$ to estimate the value of $7073^{2/3}$, giving your answer to an accuracy of 2 decimal places.

(6 marks)

Question 3

Define a real-valued function

$$f(x) = \int_{x^2-6x+12}^{x^3-6x^2+8x+4} (t^3 + \ln(t))^{1/5} \; dt$$

for $x \geq 0$.

(a) Use a non-calculator method to find the exact coordinates of the turning points of the graph

$$y = x^3 - 6x^2 + 8x + 4$$

and state the nature of those turning points. Show your working in full.

(4 marks)

(b) Explain why the functions $x^3 - 6x^2 + 8x + 4$ and $x^2 - 6x + 12$ are positive for all $x \geq 0$, and why $f(x)$ is twice differentiable on the interval $[0, \infty)$.

(3 marks)

(c) Write down an expression for $f'(x)$.

(3 marks)

(d) Show that $f''(c) = 0$ for some $c > 1$.

(5 marks)

Question 4

(a) Give an example of a bounded function $f: [-1,1] \longrightarrow \mathbb{R}$ such that f is Riemann integrable on $[-1,1]$, and both the left and right limits

$$\lim_{x \longrightarrow 0^-} f(x) \;, \qquad \lim_{x \longrightarrow 0^+} f(x)$$

do not exist.

(2 marks)

(b) Compute the limit

$$\lim_{n \longrightarrow \infty} \sum_{k=1}^{n} \frac{\sqrt{n}}{(n+k)\sqrt{n+2k}},$$

giving your answer in exact form. Justify your answer fully.

(10 marks)

Question 5

Define a function $f: [-3,5] \longrightarrow \mathbb{R}$, $f(x) = |x^2 - 2x - 3|$. A graph of the function is provided in Figure Q5 below. The graph $y = f(x)$ has a vertical line of symmetry $x = 1$, a local maximum point at $(1,4)$, and intersects the x-axis at $(-1,0)$ and $(3,0)$.

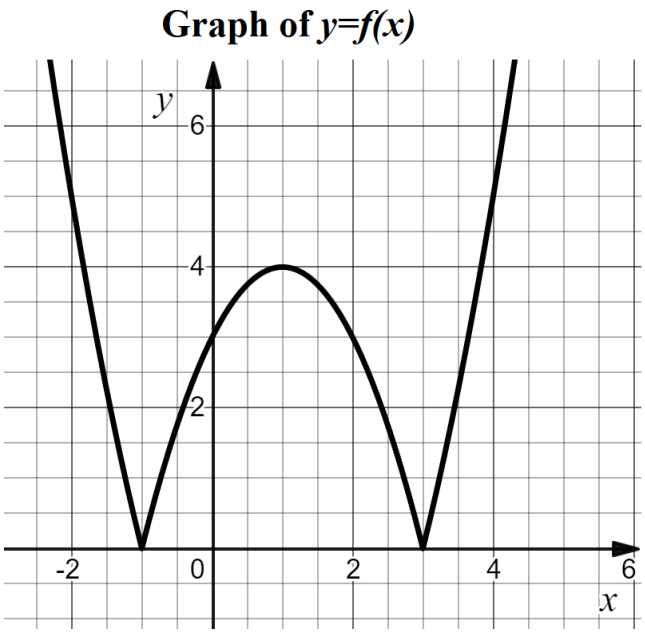

Figure Q5: Graph of $y = f(x)$

(a) Let $\mathcal{P}$ denote the tagged partition of $[-3,5]$ into 6 intervals, consisting of the intervals and tags given below:
$([-3,-2]$, tag: $-2.5)$, $([-2,-1.5]$, tag: $-1.75)$, $([-1.5,-0.5]$, tag: $-1)$,
$([-0.5,1]$, tag: $0.5)$, $([1,2]$, tag: $1.5)$, $([2,5]$, tag: $3)$.
Calculate the value of the Riemann sum $\mathcal{S}(f,\mathcal{P})$. Full working is required.

(6 marks)

(b) Let Q denote a partition of $[-3,5]$ into 8 equal-width intervals. Calculate the Lower Riemann Sum and Upper Riemann Sum with respect to Q, that is, $L(f,Q)$ and $U(f,Q)$. Full working is required.

(6 marks)

(c) Let R_{2n} denote a partition of $[-1,3]$ into $2n$ equal-width intervals. Find an expression for $U(f,R_{2n})$ in terms of n and determine the minimum value of n such that

$$\left(U(f,R_{2n}) - \int_{-1}^{3} f(x)\,dx\right) < 0.08.$$

(8 marks)

Question 6

Suppose that $f: (a, b) \longrightarrow \mathbb{R}$ is a convex function defined on the open interval (a, b), and that s, t, u are real numbers satisfying $a < s < t < u < b$.

(a) Show that if

$$\lambda = \frac{t - s}{u - s},$$

then

$$f(t) \leq (1 - \lambda) f(s) + \lambda f(u).$$

(6 marks)

(b) Show that

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s} \leq \frac{f(u) - f(t)}{u - t}.$$

(9 marks)

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